



Health Department

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File
Southwestern
Cement
Boulder

October 8, 1987

Gary A. Wilson
Southwestern Portland Cement Company
P.O. Box 529
Lyons, Colorado 80540

Re: Annual 1987 Air Pollution Inspection Report

Dear Mr. Wilson:

Enclosed is a copy of my inspection report to the Air Pollution Control Division.

The following items must be noted:

1. The fugitive emissions from the hot clinker bay are out of compliance. It is my understanding from your June 4, 1987 letter and from our discussion that corrective measures will begin in October.

2. Fugitive emissions were observed from the secondary crusher silo, the raw mill exhaust, and the gypsum silo baghouse. This equipment should be checked since no emissions are expected.

3. An Emission Permit is necessary for E Pit. Please file an application, APEN and Control Plan with the State as soon as possible.

A-Pit is grandfathered, so a Permit is not required. However, an APEN and Control Plan are necessary.

4. In 1986 the gas consumption limitations on the limestone dryer were exceeded. The company should file a revised APEN with a request for a Permit modification.

5. Fugitive emissions from the outside hopper on the south side of the A-frame may be out of compliance.

NOV 16 1987
STATIONARY SOURCES
APCD

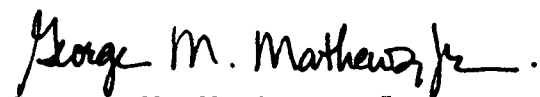
Gary A. Wilson
Southwestern Portland Cement Company
October 8, 1987
Page two

Re: Annual 1987 Air Pollution Inspection Report

6. An internal memo at the APCD indicates that the recent state test showed the kiln emissions to be out of compliance. A repeat test should be scheduled as soon as possible. I have informed Bob Jourgenson that you are waiting for a communication from him. Nevertheless, you probably should proceed anyway.

If you have any questions concerning these items, please contact me (441-1182).

Sincerely,



George M. Mathews, Jr.
Boulder County Health Department

RECEIVED

OCT 18 1987

STATIONARY SOURCES
APCD

Enclosures

xc: Air Pollution Control Division, Colorado Department of Health

GMM:hb

APCD

Key Punch Data:	County	Source	Emission	Action	Date	Agency/
	Number	Number	Point #	Type	Mo Day Yr	Indiv. ID
Line 1	101101 1 3	111013 8 12	10101 13 15	10121 16 18	11121 52	10121871 54 59 66 68
1-12 Compliance Status						

Line 2 (Dup. cols. 1-15 from Line 1)

1010151 101 101
16 18 78 80

(NOTE: Columns not shown must be left blank)
(Do not punch Line 2 if column 78 is blank)

Company: Southwestern Portland Cement Co.
Site Location: Lyons
Contact Person: Gary Wilson
Telephone No.: 534-4206

Inspected by: G. Mathews
County: Boulder
Time: 9:00 - 11:45
EIS Update: Attached
9.50 Hrs.* Permit No. C-12,444

Inspection: Routine ☒ Complaint ☐ FA ☐

Emission Points Subject to NSPS: Clinker cooler

Annual inspection per 1986 APCD Contract. Prearranged inspection.
Plant tour w/ provision of 1986 operations reports.

Emission points are reported in order (actual or proposed) of
EIS listings, not in order of process flow.

EIS Pt. 001, Clinker cooler. Clinker cooler baghouse exhaust
was 0% opacity at the time of inspection, Wilson stated that
this equipment has not been a source of problems.

The issue of the hot clinker drop bay has not been
solved but the company is still working on it. Plans include:

1. Curtains around the bay door to reduce the size of
the opening but still allow a FEL to access.

2. Close all holes existing in bay walls.

3. Relocate the drop point further back in the bay,

4. Install a telescopic, moveable sleeve on the drop chute.

5. Put the chute under negative pressure and exhaust
it through the clinker cooler baghouse.

Work should begin in October.

p1 - continued.

Southwestern Portland Cement Co., Source 3
2-29-87 inspection report continued.

ELS Point 002. Raw materials crushing. Some high opacity fug. dust was observed at the load hopper into the primary crusher at one time when the FEL delivered. This is unusual since material is large rock w/ little fines.

Fugitive emissions from the secondary crusher silo top access door were observed. An official opacity observation was not possible due to difficult access and observation points. However, emissions were $< 20\%$, but should be checked because 0% is expected and possible.

See note at end of report.

ELS Point 003. Coal handling. Baghouse exhaust point: 0% opacity.

ELS Point 004. Quarries.

- A Pit. This pit has been going since 1969. At this time company is actively mining this pit by moving toward the south.

Typical blasting information:

	<u>Overburden</u>	<u>Production rock</u>
Area:	$20' \times 20' = 400 \text{ ft}^2$	$14' \times 14' = 196 \text{ ft}^2$
Depth:	65'	36'

Moisture content is difficult to predict due to occurrence of springs + seeps in the formations.

A typical blast can range from 15,000 - 72,000 tons per event.

In 1986, a total of 1,712,378 tons were blasted.

- D Pit. This pit is being reclaimed. Waste dust is being deposited in the bottom and waste rock on the top and sides.

Southwestern Portland Cement Co., Source 3

9-29-87 report continued.

- E Pit. This is the only pit south of Hygiene Road. It is about 52 acres in size. It opened 2 yrs. ago and is expected to go 4-5 yrs.

An area, also about 50 ac (?) is to the south of E-Pit, beyond the "red gates." In this area, company just took (mined) the high outcroppings. Reclaiming is now going on & should be done next spring.

APENs w/ F.D. Control Plans have never been submitted for quarries. E-Pit, not grandfathered, has not been permitted.

ELS Point ODS. Site Fugitive Dust.

The plant site around the kiln, dryer, etc. is concrete drive- & roadways. Small vacuum cleaners sweep pads. To me, it looked dustier than in times past.

Quarry haul roads are watered as necessary. At the time of inspection, the roads were muddy as they have been during past inspections: Typical haul road data:

Silt content of road surface: "very low because road base is limestone & shale overburden"

Vehicle speed: 20-25 mph.

Vehicle weight: empty : 150 T
full : 200 T

wheels : 6

trucks : 4 (+1 standby) - 4 operating all time.

Schedule : 40 hours/week (overtime is rare)

50 minutes/hour. Schedule will vary from 8 hrs/day or 10 hrs/day but always 40 hrs/wk.

p3. continued.

Southwestern Portland Cement Co., Source 3
9-29-87 inspection report continued.

EIS Point 006. Dryer (and raw mtl's grinding).

Dryer baghouse exhaust: 0% opacity.

Total mtls dried in 1986: 658,024 Tons (< 717,000 T permit limit) — This was exceeded in 1985

* Total gas usage in 1986: 84,552,000 scf ($> 50 \times 10^6$ scf permit limit)

Total coal. usage in 1986: 1,135 Tons (< 7446 T permit limit)

Raw mill emissions were 0-5% opacity. Company should check.

ELS Point 007. Homogenizing silos. No problems noted.

EIS Point 008. A-Frame and outside clinker storage (84 B0396 F).

A-Frame : south door curtains had been removed Monday (9-28)
for repair + will be back up by Friday (10-2)

Outside clinker storage:

3 piles : 1 @ $\sim 30' \times 50' \times 8'$ (high)

1 @ ~ 30' x 50' x 8' (high)

1. (a) $\sim 30' \times 150' \times 10'$ (high)

Because plant was down for 3 months early in 1987, these stockpiles have been utilized.

Outside hopper is not enclosed, but I did not observe ^{STANDARD SUB. 1} ~~observed~~ ₁₉₅₂ dropping during inspection.

Southwestern Portland Cement Co., Source 3
9-29-87 inspection report continued.

EIS Point 009. Kiln Stack. 0% opacity at time of inspection.

Company is waiting for APCD's response on the 1987 stack test. However, co. knows they have to run stack tests again on kiln and dryer.

Production data for 1986 was provided. - attached
Recent coal analysis was provided. - attached.

Oil information: S %wt : 0.36%
BTU value: 133,000 BTU/gal.

EIS Point 010. Finished Products Silos : 0% opacity ; no problems observed.

EIS Point 011. Mill Building. Steam was observed exhausting from one of the finished product exhausts. I learned that water is injected into the ball mill to keep heat down (from mechanical grinding). Steam exhausts at ~ 230 °F.
Water is not injected into raw mill.

EIS Point 012. Silica silos.

Dust was observed coming from top of gypsum silo.

Company should check.

Silica is now purchased from Colorado Lean (?).

The Loukonen quarry is only used as occasional backup.

1607
STATIONARY SOURCES
APCD

Southwestern Portland Cement Co., Source No. 3
9-29-87 inspection report continued.

EIS Point 442 . continued. Waste dust silo.

Waste dust comes from the kiln and alkali baghouses.
When making any kind of cement, alkali baghouse collections are waste dust.

Only when making low alkali cement does the kiln baghouse collections go to the waste dust silo. Otherwise the collections go back into the kiln.

Approximately 8-9 loads of waste dust / 8-hr shift are hauled to pits for disposal (~ 1.25 mi one way).

The waste dust silo has always been a problem of fug. em.

Status: Clinker Handling : Not in compliance - hot clinker bay
Raw Materials : In compliance, but need to check secondary crusher baghouse.
Coal Handling : In compliance.
Quarries : Not in compliance. Need APEN + Control Plan for A Pit; APEN, Control Plan, + Permit for E Pit.
Site fugitive dust: In compliance
Dryer : Not in compliance : Gas limit exceeded.
Raw mill exhaust : opacity, need to check baghouse
NO_x stack test indicated problems.
Homogenizing silos : In compliance
K-Frame and outside clinker storage: In compliance, but outside hopper may not be.
Kiln : Not in compliance (Reg 3). Stack Test needs to be re-done
Finished Products Silo: In compliance.
Mill Building : In compliance.
Silica Silos : In compliance but need to check baghouse.

SEP 18 1987
STATIONARY SOURCE
APCD

OPERATIONS REPORT - LYONS PL.

MONTHLY

YEAR-TO-DATE

Drill pping	Ft: 8,991 Tons: 151,099	Hrs: 122.5	Ft/Hr: 73.4	Ft: 20,288 Tons: 346,381	Hrs: 293.5	Ft/Hr: 69.1		
CRUSH/DRYING:	TONS	744 HOURS	TONS/HR.	% OPER.	TONS	2160 HOURS	TONS/HR.	% OPEF
Hi.Cal.R#1					52,589	354.5	148.3	16.4
Hi.Cem.R#3					32,329	217.0	149.0	10.0
Lo Cem.R#3-4					18,356	123.5	148.7	5.7
Weathered R. #4					19,895	159.0	125.1	7.3
Silica					1,917	14.5	132.2	0.7
Gypsum	1,425				5,022	36.0	139.5	1.7
Iron					1,034	8.0	129.3	0.4
TOTALS	1,425	-	-	-	131,142	912.5	143.7	42.2
RAW GRIND	0	0	0	0	115,893	963.5	120.3	44.6
KILN	0	0	0	0	65,858	1146.5	57.4	53.1
FINISH MILL:								
I	9,892	111.0	89.1	14.9	31,913	334.0	95.5	15.6
I LA	0	0	0	0	0	0	0	0
II	7,516	85.5	87.9	11.5	27,926	292.5	95.5	13.5
II LA	0	0	0	0	632	7.0	90.3	0.3
II MOD	5,954	68.0	87.6	9.1	17,291	183.0	94.5	8.5
CLASS G	0	0	0	0	0	0	0	0
PRODUCT	1,237	27.5	45.0	3.7	5,790	125.0	46.3	5.8
M	0	0	0	0	1,355	24.5	55.3	1.1
ALS	24,599	292.0	84.2	39.2	84,907	966.0	87.9	44.8
Equiv. Prod:	.85x 0	+ .15x 24,599	= 3,690 Tons		.85x 65,858	+ .15x 84,907	= 68,715 Tons	
POWER:	KWH: 2,412,000	COST/KWH: .06650	KWH: 13,734,000	COST/KWH: .0561				
	COST: 160395.35	COST/EQ.TON: 43.47	COST: 771119.64	COST/EQ.TON: 11.22				
FUEL:	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON
DRYER	0	0	0	0	12,641	0	505	181,293
KILN	0	0	0	0	18,732	12,562	14,689	5,114,534
TOTALS	0	0	0	0	31,373	12,562	15,194	5,471,616
MANHOURS:	No. Hours	Hrs.Pd.	Total	Man Hrs/	No. Hours	Hrs.Pd.	Total	Man Hrs/
Hourly	88 15,407	989	16,396	4.44	41,860	4,439	46,299	.67
Staff	44 -	-	7,630	2.07	-	-	22,890	.33
TOTALS	132 -	-	24,026	6.51	-	-	69,189	1.01
SALES:	MONTHLY	YEAR-TO-DATE			SAFETY:	MONTHLY	YEAR-TO-DATE	
I	10,731	29,852			First Aid Cases	12	34	
I LA	0	0			Doctor Cases	1	3	
II	12,089	29,208			Lost Time Accidents	0	2	
II LA	0	0			Days since last lost	33	33	
"G"	0	0			time accident			
II MOD	6,482	19,032						
PI ICT	2,125	6,794						
M J	500	1,154						
Other	4	10						
TOTALS	31,931	86,050						

E. D. Cook
 E. D. Cook, Manager of Operations
 MONTH: MARCH 1986

MONTHLY

OPERATIONS REPORT - LYONS PLAI.

YEAR-TO-DATE

I.R. Drill					Ft: 32,317				
Stripping					Hrs: 479.0				
Ft: 1,329					Ft/Hr: 67.5				
Tons: 119,869					Tons: 73,347				
C HING:					4344				
H 1.R#1					TONS HOURS				
Hi.Cem.R#3					TONS/HR. % OPER.				
Lo Cem.R#3-4					TONS HOURS				
Weathered					TONS/HR. % OPER.				
R. #4					TONS HOURS				
Silica					TONS/HR. % OPER.				
Gypsum					TONS HOURS				
Iron					TONS/HR. % OPER.				
TOTALS					265,489 TONS TOTAL				
RAW GRIND					254,154				
KILN					2103.5				
FINISH MILL:					120.8				
I					58.4				
I LA					50,143				
II					542.0				
II LA					92.5				
II MOD					12.5				
Class G					23.5				
PRODUCT					75.8				
MASON					2.6				
TOTALS					7.3				
E Prod: .85x 41,627 + .15x 41,951 = 41,676 Tons					.85x 144,093 + .15x 195,447 = 151,796 Tons				
POWER:					KWH: 29,232,000				
FUEL:					COST/KWH: 0.0531				
DRYER					COST: 1,551,871.64				
KILN					COST/EQ.TON: 10.22				
TOTALS					1008 140,000 10,912				
MANHOURS:					1004 140,000 10,788				
Hourly					GAS MCF OIL/GAL COAL/TONS BTU/TON				
Staff					GAS MCF OIL/GAL COAL/TONS BTU/TON				
TOTALS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
SALES:					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
"G"					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II MOD					GAS MCF OIL/GAL COAL/TONS BTU/TON				
PRODUCT					GAS MCF OIL/GAL COAL/TONS BTU/TON				
NS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
TOTALS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
SALES:					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
"G"					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II MOD					GAS MCF OIL/GAL COAL/TONS BTU/TON				
PRODUCT					GAS MCF OIL/GAL COAL/TONS BTU/TON				
NS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
TOTALS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
SALES:					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I					GAS MCF OIL/GAL COAL/TONS BTU/TON				
I LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II LA					GAS MCF OIL/GAL COAL/TONS BTU/TON				
"G"					GAS MCF OIL/GAL COAL/TONS BTU/TON				
II MOD					GAS MCF OIL/GAL COAL/TONS BTU/TON				
PRODUCT					GAS MCF OIL/GAL COAL/TONS BTU/TON				
NS					GAS MCF OIL/GAL COAL/TONS BTU/TON				
TOTALS					GAS MCF OIL/GAL COAL/TONS BTU/TON				

SAFETY:

MONTHLY YEAR-TO-DATE

First Aid Cases

12

71

Doctor Cases

0

5 + 1* = 6

Lost Time Accidents

0

3 + 1* = 4

Days since last lost time accident

44

44

*Adjustment (Lost time occurred May 18-not reported at that time.)

E. D. Cook

Manager of Operations

MONTH: JUNE 1986 STATIONARY SOURCES

AFCD

OPERATIONS REPORT - LYONS PL/

MONTHLY

YEAR-TO-DATE

I.R. Drill ripping					Ft: 3,350 Hrs: 56.0 Ft/Hr: 60.0					Ft: 46,617 Hrs: 715.0 Ft/Hr: 65.2				
					Tons: 122,445					Tons: 1,042,455				
CRUSHING:					720					6552				
					TONS	HOURS	TONS/HR.	% OPER.		TONS	HOURS	TONS/HR.	% OPER.	
Hi. Cal. R#1					39,120	249.0	157.1	34.7		231,521	1596.0	145.1	24.4	
Hi. Cem. R#3					27,019	172.5	156.6	23.9		131,899	903.5	146.0	13.8	
Lo Cem. R#3-4					4,680	30.0	156.0	4.2		33,413	225.0	148.5	3.4	
Weathered R. #4					13,670	91.0	150.2	12.6		78,253	609.0	128.5	9.3	
Silica					0					0	475,086	TONS TOTAL		
Gypsum					1,757	14.0	125.5	1.9		8,740	70.0	124.8	1.1	
Iron					2,448	19.0	128.8	2.6		18,703	145.0	129.0	2.2	
TOTALS					939	7.5	125.2	1.1		5,844	46.0	127.0	0.7	
					89,633	583.0	153.7	81.0		508,373	3594.5	141.4	54.9	
RAW GRIND					75,923	587.5	129.2	81.6		447,993	3596.0	124.6	54.9	
KILN					46,813	698.5	67.0	97.0		263,708	4428.0	59.5	67.6	
FINISH MILL:														
I					9,684	95.5	101.4	13.3		67,917	721.5	94.1	11.0	
I LA					0					0				
II					8,157	80.0	102.0	11.1		147,462	1594.0	92.5	24.4	
II LA					20,984	265.5	79.0	36.9		46,204	604.0	76.5	9.2	
II MOD					2,080	20.5	101.5	2.8		37,503	405.0	92.6	6.2	
Class G					0					0				
ODUCT					1,849	37.5	49.3	5.2		16,784	353.5	47.5	5.4	
SON					0					5,801	101.5	57.2	1.5	
TOTALS					42,754	499.0	85.7	69.3		321,671	3779.5	85.1	57.7	
Equiv. Prod: .85x 46,813 + .15x 42,754 = 46,204 Tons										.85x 263,708 + .15x 321,671 = 272,402 Ton				
POWER:					KWH: 6,597,000 COST/KWH: 0.048					KWH: 48,249,000 COST/KWH: 0.051				
					COST: 318,107.89 COST/EQ.TON: 6.88					COST: 2,489,180.93 COST/EQ.TON: 9.14				
FUEL:					1009 140,000					1007 140,000 10,788				
					GAS MCF	OIL/GAL	COAL/TONS	BTU/TON		GAS MCF	OIL/GAL	COAL/TONS	BTU/TON	
DRYER					14,872	-	-	167,414		67,927	-	907	173,046	
KILN					141,724	288,150	-	3,916,444		449,089	1,384,468	27,220	4,676,979	
TOTALS					156,596	288,150	-	4,236,993		517,016	1,384,468	28,127	5,010,575	
MANHOURS:					No. Emp.	Hours Worked	Hrs. Pd. Not Worked	Total Man Hrs/Hours	Equiv. T.	No. Emp.	Hours Worked	Hrs. Pd. Not Worked	Total Man Hrs/Hours	Equiv. T.
Hourly					88	+ 5 Temp	13,255	2,296	15,551			135,778	16,640	152,418
Staff					38	-	-	6,589	.14		-	-	67,539	.25
TOTALS					126	+ 5 Temp	-	-	22,140	.48		-	-	219,957
SALES:					MONTHLY					YEAR-TO-DATE				
I					5,538					62,925				
I LA					0					0				
II					15,848					158,660				
II LA					16,376					32,031				
"G"					752					3,583				
MOD					2,580					38,406				
DUCT					1,311					16,506				
MASONS					666					4,902				
Other					2					51				
TOTALS					43,073					317,064				

SAFETY:		MONTHLY	YEAR-TO-DATE
First Aid Cases	RE	12	104
Doctor Cases		1	13
Lost Time Accidents		10	7
Days since last lost time accident		46	46
STATIONARY SOURCES			
APCD			
P. L. Cook (S.W.)			
Manager of Operations			
MONTH: SEPTEMBER 1986			

MONTHLY

OPERATIONS REPORT - LYONS PLANT

YEAR-TO-DATE

Drill	Ft: 7,045		Hrs: 99.5		Ft/Hr: 70.8		Ft: 65,709		Hrs: 1022.0		Ft/Hr: 64.3	
Shipping	Tons: 95,932						Tons: 1,332,131					
CRUSHING:	744						8760					
	TONS	HOURS	TONS/HR.	% OPER.	TONS	HOURS	TONS/HR.	% OPER.				
Hf. Cal. R#1	34,389	244.0	140.9	32.8	324,871	2272.0	143.0	25.9				
Hf. Cem. R#3	12,884	96.0	134.2	12.9	169,201	1180.0	143.4	13.5				
Lo Cem. R#3-4	1,720	13.0	132.3	1.7	40,986	281.0	145.9	3.2				
Weathered R. #4	24,721	227.0	108.9	30.5	122,966	1003.0	122.6	11.4				
	0				0	6580.24	TOTAL	4736 hrs				
Silica	1,630	13.0	125.4	1.7	12,510	100.5	124.5	1.1				
Gypsum	1,074	10.0	107.4	1.3	24,629	199.5	123.5	2.3				
Iron	1,045	8.5	122.9	1.1	8,877	70.5	125.9	0.8				
TOTALS	77,463	611.5	126.6	82.0	704,040	5106.5	137.9	58.3				

RAW GRIND	70,683	626.0	112.9	84.1	624,573	5126.5	121.8	58.5
KILN	37,707	666.0	56.6	89.5	366,901	6157.5	59.6	70.3
FINISH MILL:					1986 clinker prod.			
I	2,216	27.5	80.6	3.7	96,571	1035.0	93.3	11.8
I LA	0				0			
II	6,737	76.5	88.1	10.3	179,341	1947.5	92.1	22.2
II LA	7,692	102.5	75.0	13.8	74,353	958.0	77.6	10.9
II MOD	1,416	18.5	76.5	2.5	43,882	479.5	91.5	5.6
Class G	0				0			
P UCT	914	19.5	46.9	2.6	20,471	430.5	47.6	4.9
N	0				7,592	133.0	57.1	1.5
TALS	18,975	244.5	77.6	32.9	422,210	4983.5	84.7	56.9

Equiv. Prod: $.85 \times 37,707 + .15 \times 18,975 = 34,897$ Tons $.85 \times 366,901 + .15 \times 422,210 = 375,197$ Tons

POWER:	KWH: 6,129,000		COST/KWH: .0512		KWH: 66,762,000		COST/KWH: .0513	
	COST: 313,887.43		COST/EQ. TON: 9.01		COST: 3,427,874.46		COST/EQ. TON: 9.14	
FUEL: DRYER KILN TOTALS	994	135,000	10,480		1003.6	135,000	10,693	
	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON	GAS MCF	OIL/GAL	COAL/TONS	BTU/TON
	2,921	-	430	153,832	84,552	-	1,735	173,230
	12,041	421,396	5,055	4,636,010	591,549	2,387,117	37,680	4,692,721
	14,962	421,396	5,485	4,952,032	676,101	2,387,117	39,415	5,025,129

MANHOURS:	No. Emp.	Hours Worked	Hrs. Pd. Not Worked	Total Hours	Man Hrs/ Equiv. T.	No. Emp.	Hours Worked	Hrs. Pd. Not Worked	Total Hours	Man Hrs/ Equiv. T.
Hourly	89	13,984	1,839	15,823	.46		177,739	21,806	199,545	.53
Staff	39	-	-	6,763	.19		-	-	87,480	.23
TOTALS	128	-	-	22,586	.65		-	-	287,025	.76

Overtime Hours: 389.5

SALES:	MONTHLY	YEAR-TO-DATE
I	2,904	93,380
I LA	0	0
II	8,251	196,403
II LA	4,055	52,307
"G"	0	3,610
II MOD	1,755	44,703
P CT	1,301	19,679
MAINT	245	6,152
Other	19	83
TOTALS	18,530	416,317

SAFETY:	MONTHLY	YEAR-TO-DATE
First Aid Cases	15	141
Doctor Cases	1	15
Lost Time Accidents	0	8
Days since last lost time accident	90	90

E. D. Cook, Manager of Operations

MONTH: DECEMBER 1986



COMMERCIAL TESTING & ENGINEERING CO.

GENERAL OFFICES: 1919 SOUTH HIGHLAND AVE., SUITE 210-B, LOMBARD, ILLINOIS 60148 • (312) 953-9300

SINCE 1908

Member of the SGS Group (Société Générale de Surveillance)

PLEASE ADDRESS ALL CORRESPONDENCE TO:
490 ORCHARD ST., GOLDEN, CO 80401
TELEPHONE: (303) 278-9521

Southwestern Portland Cement
Mountain Division
P. O. Box 529
Lyons, Colorado 80504

Date: February 10, 1987
IAD #97-Y019-04
Received: 2/2/87
P.O. #5747

Attention: R. Trujillo

Material: Coal

Procedure: Proximate Analysis per ASTM, Part 05.05, Method D4326-84.

Ultimate Analysis per ASTM, Part 05.05, Method D3176-74.

Results: Results are reported in weight percent (Wt. %)

Coal Analysis

	Material Before Dryer		Kiln Feed 1300hr		Kiln Feed : 1300hr	
	----- 1/9/87 -----		----- 1/7/87 -----		----- 1/9/87 -----	
	<u>As Received</u>	<u>Dry</u>	<u>As Received</u>	<u>Dry</u>	<u>As Received</u>	<u>Dry</u>
Moisture %	6.97	---	6.81	---	13.94	---
Ash %	5.04	5.42	6.43	6.90	5.09	5.91
BTU/lb	11757	12638	11721	12577	10947	12720
Sulfur %	0.39	0.42	0.48	0.51	0.34	0.39
Carbon %	68.63	73.77	67.23	72.14	62.43	72.54
Hydrogen %	4.61	4.96	4.52	4.85	4.14	4.81
Nitrogen %	1.45	1.56	1.48	1.59	1.38	1.60
Oxygen %	19.88	13.87	19.86	14.01	26.62	14.75
Chlorine %	0.00	0.00	0.00	0.00	0.00	0.00

REF: 100

100 1007

STATIONARY SOURCES
ACCP

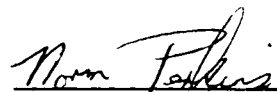
Southwestern Portland Cement
IAD #97-Y019-04
February 10, 1987
Page 2

Coal Analysis

Kiln Feed
1400 hr
----- 1/10/87 -----

	As Received	Dry
Moisture %	7.81	---
Ash %	6.45	7.00
BTU/lb	11570	12550
Sulfur %	0.41	0.45
Carbon %	66.32	71.94
Hydrogen %	4.41	4.79
Nitrogen %	1.49	1.62
Oxygen %	20.91	14.19
Cholorine %	0.01	0.01

If you have any questions regarding these results, please feel free to call.


Norm Perkins
Section Supervisor
Analytical X-Ray

NP/hn

RE

OUT 16 1987

STATIONARY SOURCES
AFCD

COMMERCIAL TESTING & ENGINEERING CO.



AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HRS

EIS POINT ID: 01
 POINT NAME: CLINKER COOLER
 HSPS REG?: YES: 6,IV,C,2,a & b
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

OP'NAL SCHED:

KILN FEED: 624,573 TONS
 CEMENT PRODUCED: 375,197 TONS
 24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	6,IV,C,2,a	6,IV,C,2,b					
DESIGN RATE							
HR ALLOW EM			10%				
ALLOW EM (LB/TON FEED TO KILN)	0.1						

YR THRUPUT (TONS CEMENT PROD)	375197
REFERENCE	3-05-006-14
EM FACTOR (LB/TON)	9.2
CONT FACTOR	0.999
YR ACTUAL EM (TONS)	1.726

KILN FEED (TONS)	624573
ACTUAL EM (LB/TON FEED TO KILN)	0.006

HR ACTUAL EM	0%
--------------	----

COMPLIANCE (3=YES;1=NO)	3	YES
REV APEND		YES

COMMENTS:

1. FILE NAME: 3_1_86
2. DUMPING OF HOT CLINKER STILL A PROBLEM; SEE INSPECTION NARRATIVE

REC-11
OCT 16 1987
STATIONARY SOURCES
APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 FIS: 2.00 HRS

EIS POINT ID: UNASSIGNED, 02
 POINT NAME: RAW MTL PRMRY CRUSHER
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIPMENT: BAGHOUSES
 658,024 TONS
 4736 HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,6	1,II,A,1					
DESIGN RATE (TONS/HR)	200						
HR ALLOW EM (LB/HR)	40.407	20%					
YR THRUPUT (TONS)	658024						
REFERENCE	3-05-006-09	INSP					
EM FACTOR (LB/TON)	0.5						
CONT FACTOR	0.999						
YR ACTUAL EM (TONS)	0.165						
TIME FACTOR (HR/YR)	4736						
HR ACTUAL EM (LB/HR)	0.069	UNKNOWN					
COMPLIANCE (3=YES;1=NO)	3	UNKNOWN					
REV APEN?	YES						

EIS POINT ID: UNASSIGNED, 02
 POINT NAME: RAW MTL SECONDARY CRUSHER
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIPMENT: BAGHOUSES
 658,024 TONS
 4736 HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,6	1,II,A,1					
DESIGN RATE (TONS/HR)	200						
HR ALLOW EM (LB/HR)	40.407	20%					
YR THRUPUT (TONS)	658024						
REFERENCE	3-05-006-09	INSP					
EM FACTOR (LB/TON)	1.5						
CONT FACTOR	0.999						
YR ACTUAL EM (TONS)	0.494						
TIME FACTOR (HR/YR)	4736						
HR ACTUAL EM (LB/HR)	0.208	0-5%					

REC-16017
 6-18-87
 STATIONARY SOURCES
 APCD

COMPLIANCE (3=YES;1=NO)

3

YES

REV APEN?

YES

POINT NAME: RAW MATERIALS TRANSFER

PERMIT: GRANDFATHERED

CONTROL EQUIP: TOTALLY ENCLOSED

W/ NEG PRESSURE TO BAGHOUSE

1986 ANNUAL THRUPUT:

658024 TON/YR

OP'NAL SCHED:

5761 HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	NO SPEC	1,11,A,1					
HR ALLOW EM		20%					
YR THRUPUT (TONS)	658024						
REFERENCE	3-05-006-12						
EM FACTOR (LB/TON)	0.3						
CONT FACTOR	0.99						
YR ACTUAL EM (TONS)	0.987						
HR ACTUAL EM		0-5%					

COMPLIANCE

NA

YES

YR ACTUAL EM TOTAL (TONS)

1.645

COMMENTS:

1. FILE NAME: 3_2 86

REF

6-16-87

STATIONARY SOURCES
APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: EIDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HRS

EIS POINT ID: UNASSIGNED, S
 POINT NAME: COAL HANDLING
 NSPS REG?: NO
 MESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:
 OPT'AL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

COAL BURNED: 39,415 TONS
 24HRS/7DAY/47WKS/YR=789%HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,a	1,II,A,1					
DESIGN RATE (TON/HR)	13						
HR ALLOW EM (LB/HR)	17,609	20%					
YR THRUPUT (TONS)	39415						
REFERENCE	3-05-010-10	NO INSP					
EM FACTOR (LB/TON)	0.02						
CONT FACTOR	0.999						
YR ACTUAL EM (TONS)	.000						
TIME FACTOR (HR/YR)	789%						
HR ACTUAL EM (LB/HR)	.000	UNKNOWN					
COMPLIANCE (3=YES;1=NO)	3	UNKNOWN					
REV APEN?	NO						

COMMENTS:
 1. FILE NAME: 3_3_86

REC-1
 16 1037
 STATIONARY SOURCES
 APCD

ATR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HR

EIS POINT ID: 4
 POINT NAME: GUARTEES
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: NONE

ROCK BLASTED: 1,712,370 TONS
 24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	NO SPEC	1,III,2,d					
DESIGN RATE							
HR ALLOW EM		20% and					
		NO OFFPROP					

YR THRUPUT (TONS)	1712370	
REFERENCE	3-05-020-09	INSP
EM FACTOR (LB/TON)	0.16	
CONT FACTOR	0	
YR ACTUAL EM (TONS)	136.990	

HR ACTUAL EM	UNKNOWN
--------------	---------

COMPLIANCE	NA	UNKNOWN
REV APEN?		NO

COMMENTS:
 1. FILE NAME: 3_4_86

RECEIVED
 SEP 30 1987
 STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00_11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 ETS: 2.00 HRS

ETS POINT ID: 05
 POINT NAME: SITE FUGITIVE DUST
 1986 ANNUAL THRUPUT:
 OP'NAL SCHED:

PERMIT: NONE
 CONTROL: FREQ WATER
 176,250 VMT
 3HRS/SDAY/47WKS/YR=2000HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	NO SPEC 1,III,D,2,e						
DESIGN RATE							
HR ALLOW EM	NO OFFPROP						
YR ALLOW EM							

YR THRUPUT (VMT)	176250
REFERENCE	APCD FD MEMO INSP
	SECT V1
	11.2.1-2.

EM FACTOR (LB/VMT)	
k (PART SIZE)	0.8
s (SILT %)	3
S (SPEED)	22.5
W (TRUCK WT)	100
w (# WHEELS)	6
p (# WET DAYS)	82
	9.764
CONT FACTOR	0.5
YR ACTUAL EM (TON)	431.089

HR ACT EM	NO OFFPROP
-----------	------------

COMPLIANCE	NA	YES
REV APEN?		

COMMENTS:

1. FILE NAME: 3_5_86
2. $VMT = (22.5 \text{ MI/HR})(40 \text{ HR/WK})(5/6)(47 \text{ WK/YR})(5 \text{ TRUCKS}) = 176,250 \text{ VMT}$
3. CONTROL EFFICIENCY MAY BE >50%.

REC

OCT 16 1987

STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDG 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 ETS: 2.00 HRS

ETS POINT ID: 06
 POINT NAME: LIMESTONE DRYER
 (AND RAW MTLG GRINDING)
 HSPS REG?: YES: 6, R, JV
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:

PERMIT: C-12,444(1) MOD
 CONTROL EQUIP: BAGHOUSE

CEMENT PRODUCED: 375,197 TONS
 GAS BURNED: 84,552,000 SCF
 #2011 BURNED: 0
 COAL BURNED: 1,735 TONS
 4736 HR/YR

OP'NAL SCHED:

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	C-12,444(1)	1, II, A, 1	NO SPEC	C-12,444(1)	C-12,444(1)	C-12,444(1)	NO SPEC
DESIGN RATE (TON/HR)	150						
YR LIMESTONE LIMIT (TON)	717000						
YR COAL LIMIT (TON)	7446						
YR NAT GAS LIMIT (10E6 SCF)	50						
HR ALLOW EM (LB/HR)	13.9	20%		0.9	23.3	23.9	
YR ALLOW EM (TON/YR)	33.2			2.23	55.8	57.3	
ALLOW EM (LB/10E6 BTU)						1.2	
YR THRUPUT LIMESTONE (TON)	650024						
YR THRUPUT (TONS CEMENT PROD)	375197						
REFERENCE	3-05-006-13	INSP					
EM FACTOR (LB/TON)	64						
CONT FACTOR	0.999						
YR ACTUAL EM (TON)	12.006						
YR THRUPUT (10E6 SCF GAS BURNED)	84.552		84.552	84.552	84.552	84.552	
REFERENCE	1-02-006-01		SAME	SAME	SAME	SAME	
EM FACTOR (LB/10E6 SCF)	3		1.4	40	550	0.6	
CONT FACTOR	0.999		0	0	0	0	
YR ACTUAL EM (TON)	.000		0.059	1.691	23.252	0.025	
BTU VALUE (BTU/SCF)						991	
YR HEAT INPUT (10E6 BTU/YR)						83791.032	
YR THRUPUT (TON COAL BURNED)	1735		1735	1735	1735	1735	
REFERENCE	3-90-002-01		SAME	SAME	SAME	SAME	
EM FACTOR (LB/TON)	0		0	0	0	26	
% SULFUR							
CONT FACTOR	0.999		0	0	0	0	
YR ACTUAL EM (TON)	0.000		0.000	0.000	0.000	9.022	

RE 83791.032

STATIONARY SOURCES
 APCD

BTU VALUE (BTU/TON)					24	
YR HEAT INPUT (10E6 BTU/YR)					41640	
YR ACTUAL EM TOTAL (TONS)	12.006	0.059	1.691	23.252	0.025	
ACT EM (LB/10E6 BTU)					0.001	
TIME FACTOR (HR/YR)	5653					
HR ACTUAL EM (LB/HR)	4.248	0%				
COMPLIANCE (3=YES;1=NO)						
YR LIMESTONE LIMIT	3					
YR COAL LIMIT	3					
YR NAT GAS LIMIT	1					
HR	3	YES	3	3	3	NA
YR	3		3	3	3	
LB/BTU					3	
REV APEND (3=YES;1=NO)	YES					
REFERENCE	3,II,B.2,b,ii					

COMMENTS:

1. FILE NAME: 3_6_86
2. EM FACTORS FOR PART, VOC, CO, AND NOX FOR FUELS
ARE NOT PROVIDED IN SCC CODE BOOK.

REF (1)

16 1037

STATIONARY SOURCES
APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SM PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9_29_87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HRS

EIS POINT ID: UNASSIGNED
 POINT NAME: HOMOGENIZING SILOS
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPTUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

704,040 TON MTL
 24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,b	1,II,A,1					
DESIGN RATE (TON/HR)		120					
HR ALLOW EM (LB/HR)	37.236	20%					

YR THRUPTUT
 REFERENCE
 EM FACTOR
 CONT FACTOR
 YR ACT EM
 TIME FACTOR
 HR ACT EM

NONE ID'D

INSP

0%

COMPLIANCE
 REV APEN?

YES

COMMENTS:
 1. FILE NAME: 3_7_86

RECEIVED

16 1987

STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HRS

EIS POINT ID: UNASSIGNED, 00
 POINT NAME: A-FRAME AND
 CLINKER STORAGE
 1986 ANNUAL THRUPUT:

PERMIT: A-FRAME: NONE
 OUTSIDE: 84B0369F
 CONTROL EQUIP: NONE
 A-FRAME: 30,000 TONS
 OUTSIDE: 40,000 TONS
 24HRS/7DAY/52WKS/YR=8760HR/YR

OP'NAL SCHED:

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	84B0369F	1,II,A,1					
		1,III,D,2,c					

LOADOUT DESIGN RATE (TON/HR)	100
MAX OUTSIDE STORAGE (TON)	120000
HR ALLOW EM (LB/HR)	12.33
	20% & NO OFF PROP
YR ALLOW EM (TON/YR)	10.2

OUTSIDE PILES	
YR THRUPUT (TON)	40000
REFERENCE	INSP
EM FACTOR (LB/TON)	0.1
CONT FACTOR	0
YR ACTUAL EM (TON/YR)	2
TIME FACTOR (LB/YR)	8760
HR ACTUAL EM (LB/HR)	0.457
	(20% & NO OFF PROP

COMPLIANCE (3=YES;1=NO)	
OUTSIDE STORAGE LIMIT	3
HR	3
YR	3
	NO
	YES

REV APEN?

COMMENTS:

1. FILE NAME: 3_8_86
2. TRANSFER OF MTL BY FRONTLOADER INTO OUTSIDE HOPPER NOT OBSERVED

RE

16 1987

STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HR
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 ETS: 2.00 HRS

EIS POINT ID: 09
 POINT NAME: KILN STACK
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:

PERMIT: C-12,444(2)
 CONTROL EQUIP: BAGHOUSE

KILN FEED: 624,573 TONS
 CEMENT PRODUCED: 375,197 TONS
 COAL BURNED: 37,680 TONS (0.4% SULFUR; 6% ASH; 12,000 BTU/LB)
 #2 OIL BURNED: 2,587,117 GAL (0.36% SULFUR); 133,000 BTU/GAL
 GAS BURNED: 591,549,000 SCF (991 BTU/SCF)
 6157.5 HR/YR

OPERATIONAL SCHEDULE:

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	C-12,444(2)	1,11,A,1	NO SPEC	NO SPEC	NO SPEC	C-12,444(2)	
DESIGN RATE (TON/HR)	102						
HR ALLOW EM (LB/HR)	27	20%					
YR ALLOW EM (TON/YR)	102					851	
ALLOW EM (LB/TON FEED TO KILN)	0.3						
ALLOW EM (LB/10E6 BTU)						1.2	

YR THRUPUT (TONS OF CEM PROD)	375197				375197	375197
REFERENCE	3-05-006-06	INSP			3-05-006-06	3-05-006-06
EM FACTOR (LB/TON)	245		0	0	2.6	10.2
CONT FACTOR	0.999				0	0.75
YR ACTUAL EM (TONS)	45.962				487.756	478.376

TIME FACTOR (HRS/YR)	6157.5
HR ACTUAL EM (LB/HR)	14.929
	0%

KILN FEED (TON)	624573
ACTUAL EM (LB/TON FEED TO KILN)	0.147

YR THRUPUT (TONS COAL BURNED)
 REFERENCE
 EM FACTOR (LB/TON)
 %S
 CONT FACTOR
 YR ACTUAL EM (TON)
 BTU VALUE (10E6 BTU/TON)
 YR HEAT INPUT (10E6 BTU)

RE
 37680
 3-90-002-01
 26
 0.4
 16
 48,984
 STATIONARY SOURCES
 APCD
 904320

YR THRUPUT (10E3 GAL BURNED)	2387.117
REFERENCE	3-90-004-02
EM FACTOR (LB/10E3 GAL)	108

BS			0.36
CONT FACTOR			0.75
YR ACTUAL EM (TON)			11.601
BTU VALUE (10E6 BTU/GAL)			0.133
YR HEAT INPUT (10E6 BTU)			317486.561

YR THRUPUT (10E6 SCF BURNED)			591.549
BTU VALUE (BTU/SCF)			991
YR HEAT INPUT (10E6 BTU)			506225.059

YR ACT EM TOTAL (TONS)	45.962	407.756	538.962
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YR HEAT INPUT TOTAL (10E6 BTU)			1808031.62
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YR ACTUAL EM (LB/10E6 BTU)			0.596
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COMPLIANCE (3=YES;1=NO) YR	3	YES	3
HR	3		NA
FEED	3		NA
BTU	NA		3

REV APEN? (3=YES;1=NO)	1		3
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REFERENCE	3,II,B,2,b,ii		3,II,B,2,b,iv
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COMMENTS:

1. FILE NAME: 3_9_85
2. EM FACTORS FOR PART, NOX, VOC, AND CO FOR FUELS ARE NOT PROVIDED IN ACC CODE BOOK.

REC-16-1037

16-1037

STATIONARY SOURCES
APCD

ATR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-29-87
 TIME: 9:00-11:45AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 ETS: 2.00 HRS

EIS POINT ID: 10
 POINT NAME: FINISHED PRODUCTS
 SILOS

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPUT:

TOTAL CEMENT PRODUCED: 375,197 TONS

OP'NAL SCHED:

24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LFAD
REGULATION	I,III,C,I,b	I,II,A,I					
DESIGN RATE (TON/HR)	85						
HR ALLOW EM (LB/HR)	35.237	20%					
YR THRUPUT (TONS)	375197						
REFERENCE	3-05-006-19	INSP					
EM FACTOR (LB/TON)	0.24						
CONT FACTOR	0.999						
YR ACTUAL EM (TONS)	0.045						
TIME FACTOR (LB/YR)	7896						
HR ACTUAL EM (LB/HR)	0.011	0%					
COMPLIANCE (3=YES;1=NO)	3	YES					
REV APEN?	NO						

COMMENTS:
 1. FILE NAME: 3_10_86

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STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LYONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4206
 DATE: 9-2-87
 TIME: 9:00-11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 ETS: 2.00 HRS

ETS POINT ID: 11
 POINT NAME: MILL BUILDING
 NSPS REG?: NO
 HESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPTUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

TOTAL CEMENT PRODUCED: 375,197 TONS
 24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	OPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,b	1,II,A,1					
DESIGN RATE (TON/HR)	85						
HR ALLOW EM (LB/HR)	35.237	20%					
YR THRUPTUT (TONS)	375197						
REFERENCE	3-05-006-17	INSP					
EM FACTOR (LB/TON)	96						
CONT FACTOR	0.999						
YR ACTUAL EM (TONS)	18.009						
TIME FACTOR (HR/YR)	7896						
HR ACTUAL EM (LB/HR)	4.562	0%					
COMPLIANCE (3=YES;1=NO)	3	YES					
REV APEN?	NO						

COMMENTS:
 1. FILE NAME: 3_11_86

REF: 16 107

16 107

STATIONARY SOURCES
 APCD

AIR POLLUTION CONTROL COMPLIANCE DETERMINATION WORKSHEET

SOURCE NUMBER: 3
 SOURCE NAME: SW PORTLAND CEMENT CO
 LOCATION: LIONS
 COUNTY: BLDR 0220
 CONTACT: GARY WILSON
 TELEPHONE: 534-4204
 DATE: 9-29-07
 TIME: 9:00_11:45 AM

INSPECT TYPE: ANNUAL
 ACTION TYPE: 12
 INSPECTOR: G. MATHEWS
 HRS REQUIRED: 9.50 HRS
 PREPARATION: 1.00 HRS
 TRAVEL: 1.00 HRS
 INSPECTION: 2.75 HRS
 WRITE UP: 4.75 HRS
 EIS: 2.00 HRS

EIS POINT ID: UNASSIGNED,13
 POINT NAME: SILICA SILOS
 NSPS REG?: NO
 NESHAPS REG?: NO
 PSD REG?: NO
 1986 ANNUAL THRUPTUT:
 OP'NAL SCHED:

PERMIT: GRANDFATHERED
 CONTROL EQUIP: BAGHOUSE

46,016 TONS
 24HRS/7DAY/47WKS/YR=7896HR/YR

POLLUTANT	PART	CAPACITY	VOC	CO	NOX	SOX	LEAD
REGULATION	1,III,C,1,a	1,II,A,1					
DESIGN RATE (TONS/HR)	4						
HR ALLOW EM (LB/HR)	8.480	20%					
YR THRUPTUT (TONS)	46016						
REFERENCE		INSP					
EM FACTOR	NONE TD'D						
CONT FACTOR							
YR ACTUAL EM (TONS)							
TIME FACTOR (HR/YR)							
HR ACTUAL EM (LB/HR)		0-10%					
COMPLIANCE	UNKNOWN	YES					
REV APEN?		NO					

COMMENTS
 1. FILE NAME: 3_12_86

REF 16 1337

STATIONARY SOURCES
 APCD

SW PORTLAND CEMENT CO., SOURCE 3
1986 EMISSION INVENTORY

FIS FT ID	NAME	POLLUTANT (TONS)		VOC	CO	NOX	LEAD
		PART	SOX				
01	CLINKER COOLER	1.726					
02 (UNASSIGNED)	RAW MTL CRUSHERS	1.645					
03 (UNASSIGNED)	COAL HANDLING	0.000					
04	QUARRIES	136.990					
05	SITE FUG DUST	431.089					
06	LIMESTONE DRYER	12.006	0.025	0.059	1.691	23.252	
07 (UNASSIGNED)	HOMOGENIZING SILOS						
08 (UNASSIGNED)	A-FRAME & CLINKER STORAGE	2.000					
09	KILN STACK	45.962	538.962			487.756	
10	FIN PRODUCTS SILOS	0.045					
11	MILL BUILDING	18.009					
12 (UNASSIGNED)	SILICA SILOS						
	TOTAL	649.472	538.987	0.059	1.691	511.008	

COMMENTS:

1. FILE NAME: 3_86

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JUL 16 1987

STATIONARY SOURCES
APCD